

Personal Project

Portfolio

Global Electronics Retailer

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BACKGROUND

This project analyzes sales performance of a fictitious global electronics retailer operating across multiple countries, sales channels, and currencies. The dataset simulates real-world retail operations to provide insights into revenue performance, customer behavior, and market dynamics.

The analysis aims to evaluate overall business performance and identify strategic opportunities for growth and optimization.

<https://github.com/saniarizka/global-electronics>

Data Structure Overview

The dataset consists of transactional and dimensional tables representing key components of retail operations.

Stores

- store_key
- country
- state
- square_meters
- open_date

Sales

- order_number
- line_item
- order_date
- delivery_date
- customer_key
- product_key
- store_key
- quantity
- currency_code

Products

- product_key
- product_name
- brand
- color
- category
- category_key
- subcategory
- subcategory_key
- unit_cost_usd
- unit_price_usd

Customers

- customer_key
- name
- gender
- birthday
- state
- state_code
- city
- country
- continent
- zip_code

Exchange Rate

- currency
- date
- exchange

Analytical Framework

A structured approach to deliver reliable and actionable insights.

01

Business Understanding

Define business objectives, identify key metrics, and align analysis with strategic goals.

02

Data Understanding

Explore dataset structure, assess data quality, and validate data relevance.

03

Data Preparation

Clean, transform, and engineer data to ensure accuracy and analytical readiness.

04

Data Analysis & Modeling

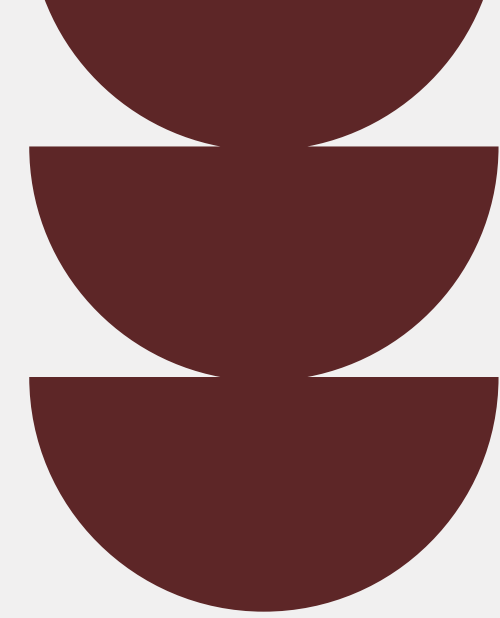
Perform structured queries and aggregations to uncover performance drivers and trends.

05

Evaluation & Visualization

Validate insights, develop dashboards, and translate findings into actionable recommendations.

Business Understanding



This analysis aims to evaluate sales performance and identify key drivers of revenue, profitability, and operational efficiency.

The analysis focuses on answering the following key questions:

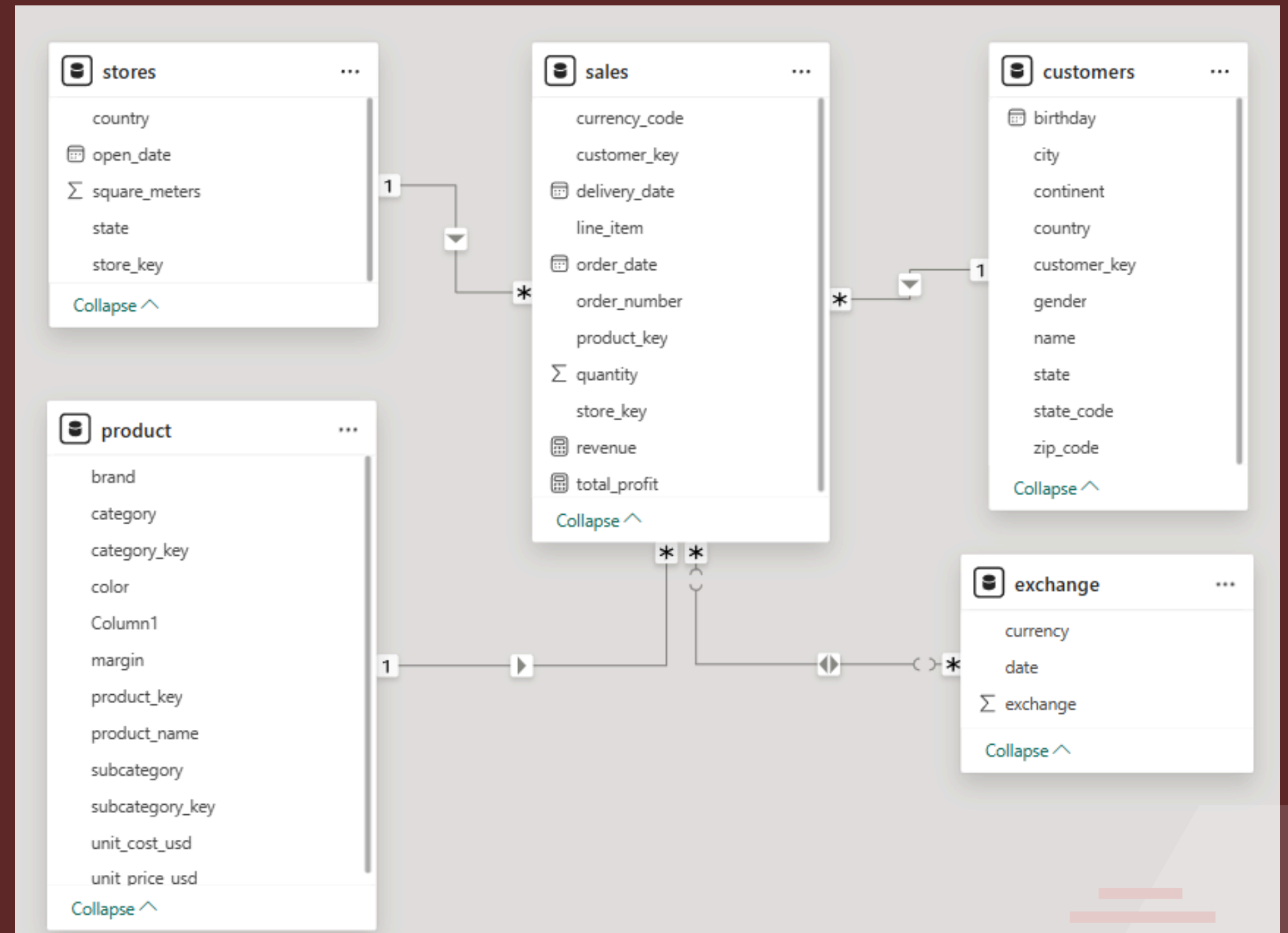
- How have revenue and profit evolved over time, and which product categories drive overall performance?
- How does sales performance evolve over time, including seasonal trends and patterns?
- Which customer age segments drive the highest transaction volume?
- Which sales channel contributes most effectively to overall performance?
- How are customers and sales distributed geographically?



Data Understanding

Here is the relational database structure of the dataset.

The model integrates all entities through structured key relationships, providing a solid foundation for cross-functional sales and customer analysis.



Data Preparation

The initial stage involved preparing a dataset consisting of five relational tables. The objective was to improve data quality and consistency prior to analysis by identifying and handling missing values, removing duplicate records, standardizing data formats (dates, currency, and categorical variables), and converting data types to ensure computational accuracy.

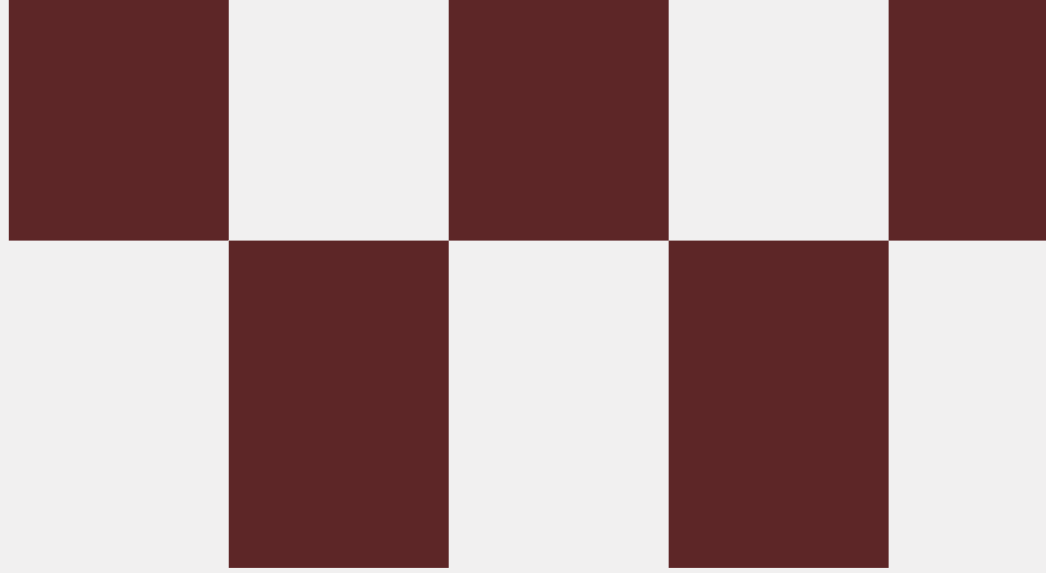
→ STORE TABLE

- The dataset contains 67 distinct entries with no duplicates identified.
- One missing value was found in the square meters variable, associated with the online store category.
- The missing value is logically justified and does not require imputation, since online businesses do not operate in physical spaces.

	store_key	country	state	square_meters	open_date
0	1	Australia	Australian Capital Territory	595	2008-01-01
1	2	Australia	Northern Territory	665	2008-01-12
2	3	Australia	South Australia	2000	2012-01-07
3	4	Australia	Tasmania	2000	2010-01-01
4	5	Australia	Victoria	2000	2015-12-09
...	...	...	...	...	...
62	63	United States	Utah	2000	2008-03-06
63	64	United States	Washington DC	1330	2010-01-01
64	65	United States	West Virginia	1785	2012-01-01
65	66	United States	Wyoming	840	2014-01-01
66	0	Online	Online	0	2010-01-01

67 rows × 5 columns

Data Preparation



➔ SALES TABLE

- The sales dataset covers the period from 01-01-2016 to 20-02-2021, consisting of 62,884 transactions.
- A total of 49,719 values (approximately 79%) are missing in the delivery_date column. Distribution analysis by sales channel shows that all online transactions have missing delivery dates.

➔ CUSTOMERS TABLE

- The customer dataset contains 15,266 unique customers distributed across 3 continents and 8 countries, with the majority located in North America.
- Geographic fields were reviewed to ensure consistency between state and state_code. A missing value in the state_code column (Napoli entry) was handled by introducing a new category ("NA") to maintain categorical consistency.
- Categorical variables were standardized, and the birthday column was converted to datetime format to enable demographic and age-based analysis.

Data Preparation

→ PRODUCTS TABLE

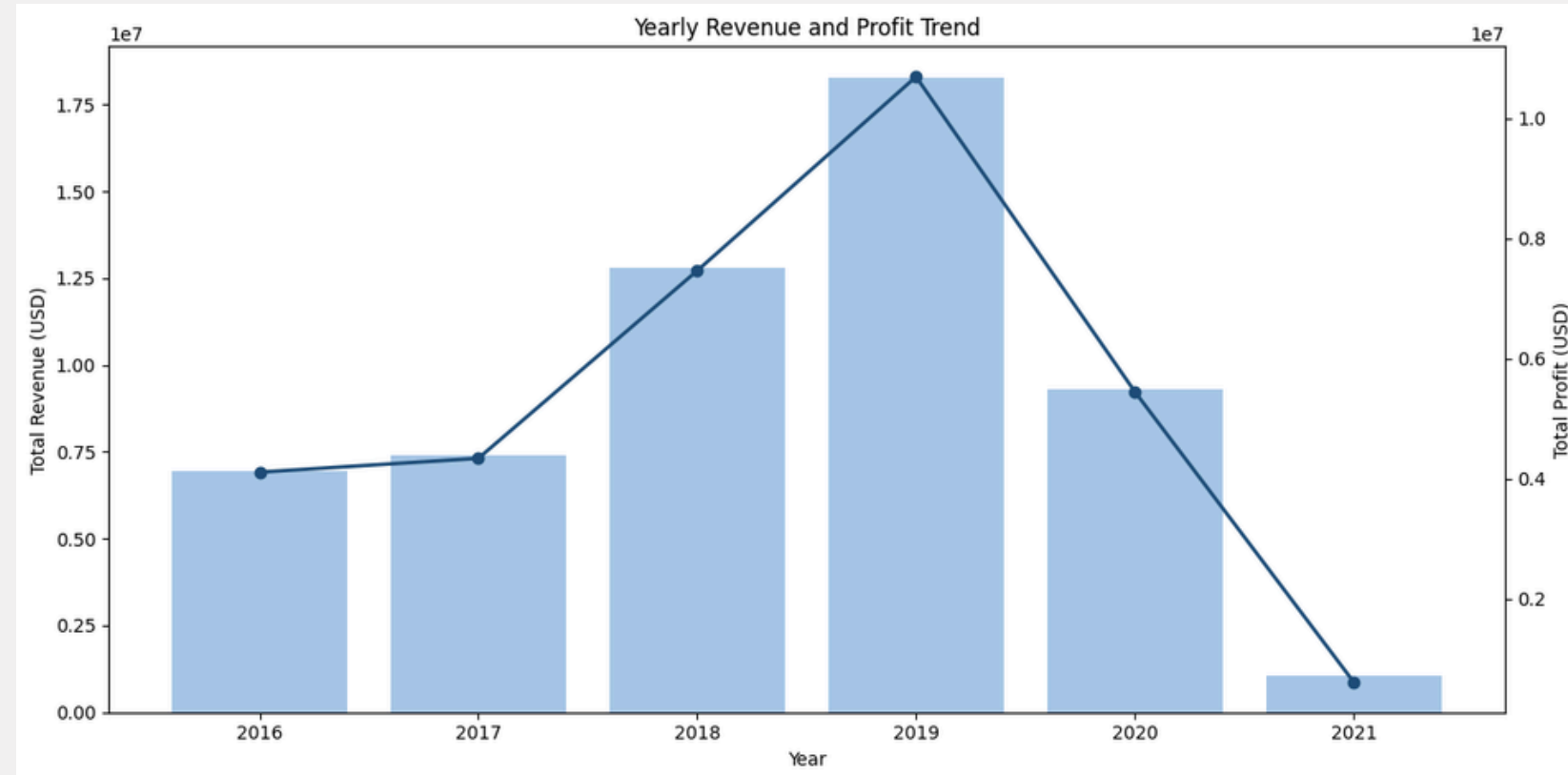
- The product dataset consists of 2,517 unique products, with no duplicate records identified.
- Products are organized into a structured hierarchy of 8 main categories, 32 subcategories, and 11 brands, enabling analysis at multiple aggregation levels.
- To ensure accurate financial calculations, currency symbols were removed from the unit_cost_usd and unit_price_usd columns, and both fields were converted into numeric (float) format. This preparation enables precise revenue, margin, and profitability analysis.

→ EXCHANGE RATES TABLE

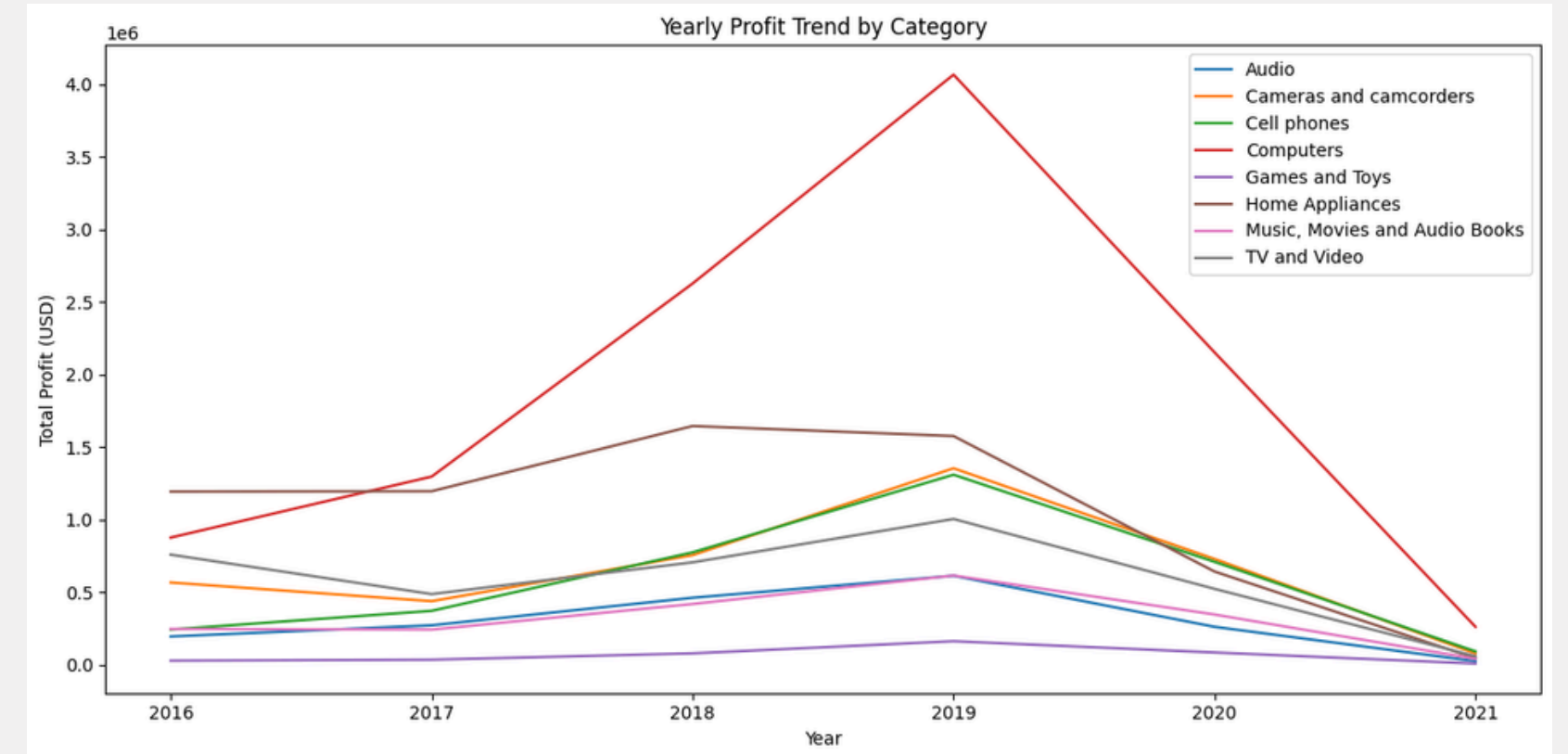
The exchange rates table was prepared to support currency normalization across different markets. Date fields were standardized, and exchange rate values were validated in numeric format to ensure accurate currency conversion for financial analysis.

	Date	Currency	Exchange
0	1/1/2015	USD	1.0000
1	1/1/2015	CAD	1.1583
2	1/1/2015	AUD	1.2214
3	1/1/2015	EUR	0.8237
4	1/1/2015	GBP	0.6415

How have revenue and profit evolved over time, and which product categories contributed most to their growth?



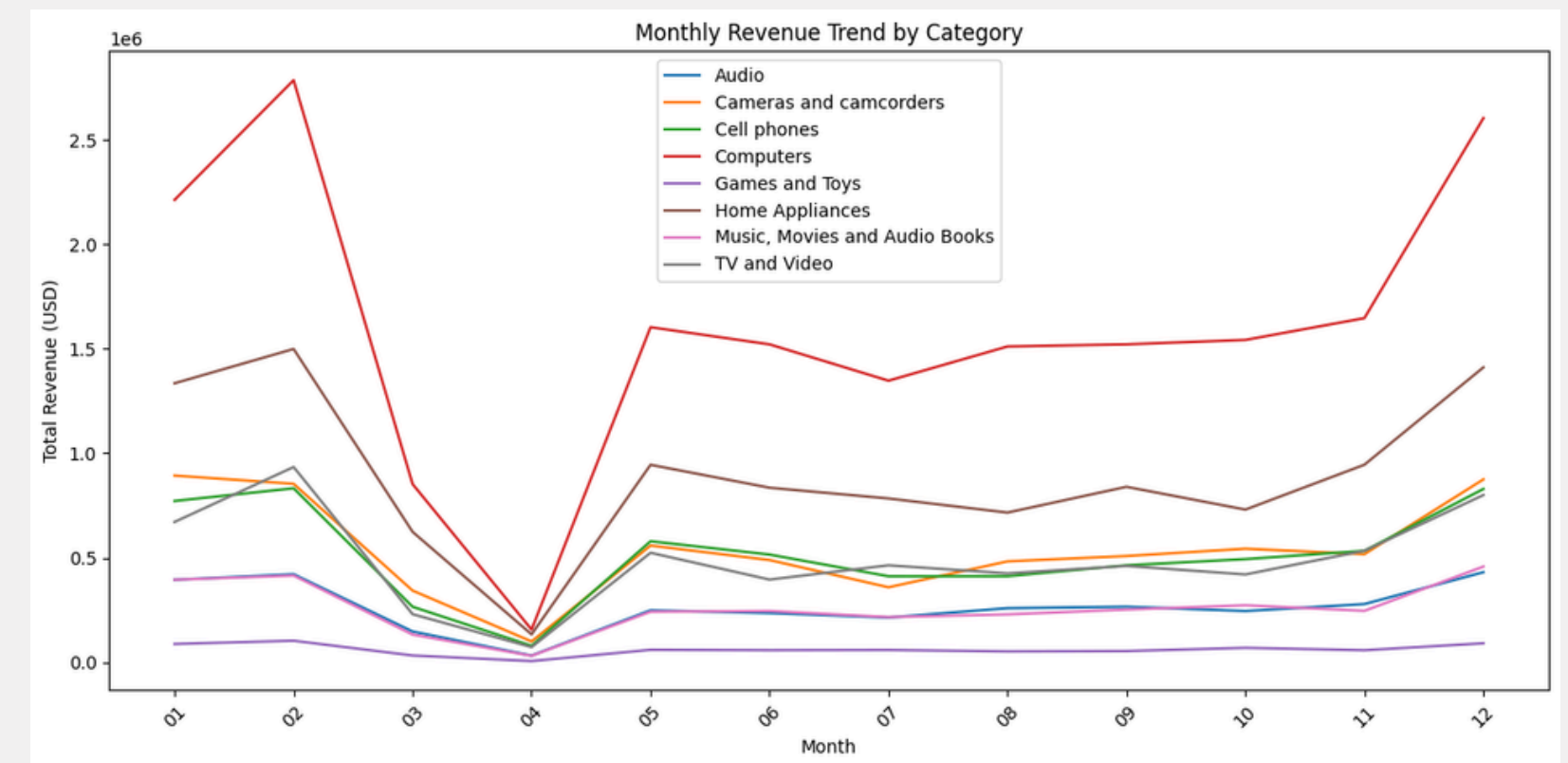
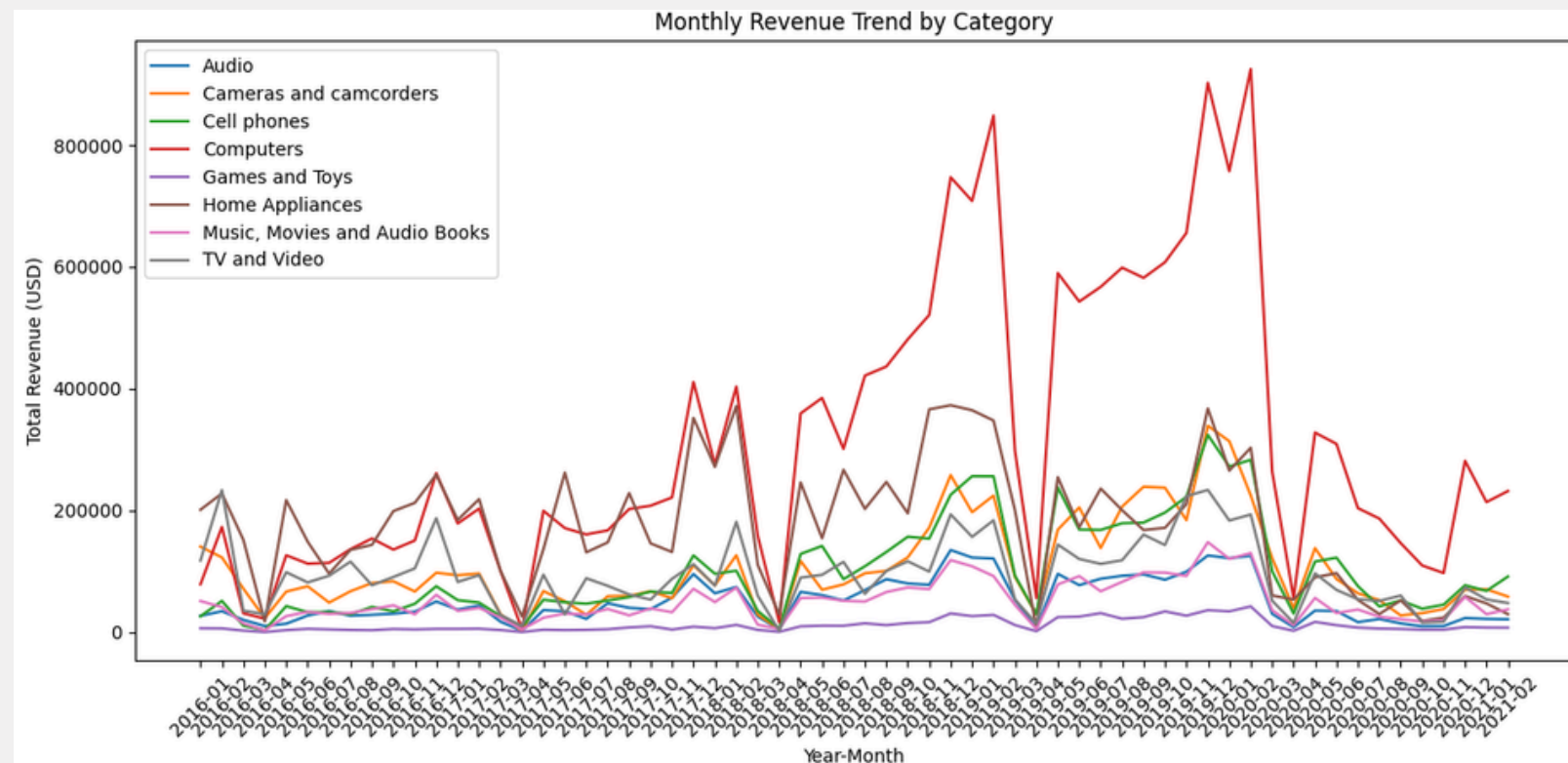
- Visually, revenue and profit move consistently in the same direction over time, showing strong alignment between sales growth and overall profitability.
- A significant upward trend is observed during 2017–2019, followed by a sharper and more pronounced decline from 2019–2021.



To better understand what drives this pattern, the analysis was extended to product-category level.

- The Computers category consistently delivered the highest profit, making it the primary profit driver of the business.
- All categories experienced steady growth from 2017–2019, reflecting a broad expansion phase.
- From 2019–2021, revenue and profit declined across categories, suggesting external or macro-level market pressures rather than category-specific issues.

How does sales performance evolve over time, including seasonal trends and patterns?



- A clear seasonal pattern is observed, with recurring peaks toward year-end and noticeable dips around March–April.
- The Computers category acts as the primary growth driver, contributing the largest share of revenue and profit.
- Due to its substantial contribution, fluctuations in this category heavily influence overall business performance and introduce higher volatility.

Which customer age segments drive the highest transaction volume?

```
WITH base_data AS (  
  SELECT  
    s.customer_key, s.quantity, p.unit_price_usd,  
    (s.quantity * p.unit_price_usd) AS revenue,  
    CAST((julianday(s.order_date) - julianday(c.birthday)) / 365.25 AS INT) AS age  
  FROM sales s  
  JOIN customers c ON s.customer_key = c.customer_key  
  JOIN products p ON s.product_key = p.product_key  
)  
  
age_category AS (  
  SELECT  
    CASE  
      WHEN age BETWEEN 12 AND 17 THEN '12-17'  
      WHEN age BETWEEN 18 AND 25 THEN '18-25'  
      WHEN age BETWEEN 26 AND 40 THEN '26-40'  
      WHEN age BETWEEN 41 AND 60 THEN '41-60'  
      ELSE '>60'  
    END AS age_group,  
    revenue, customer_key  
  FROM base_data  
)
```

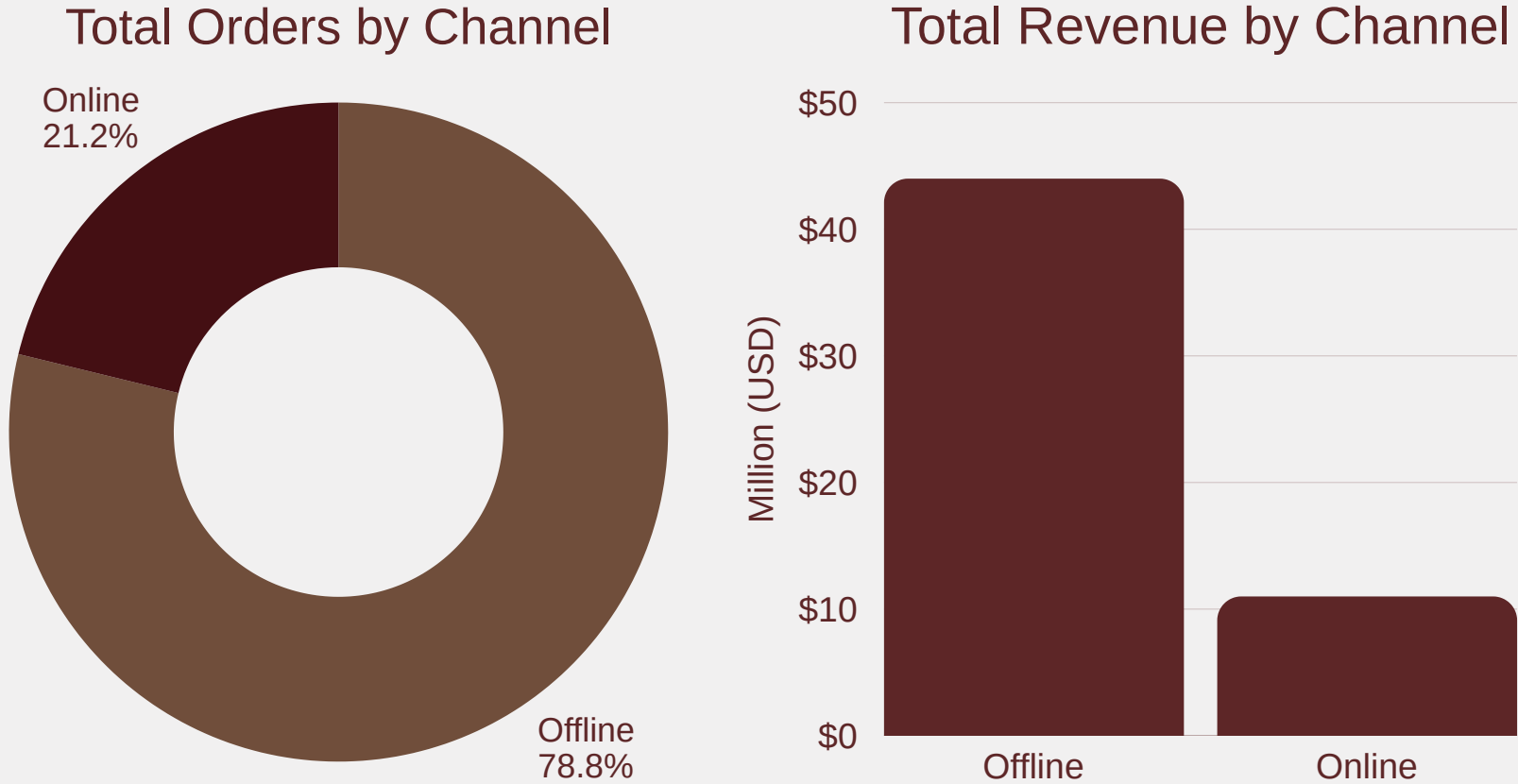
```
SELECT  
  age_group, COUNT(*) AS total_orders,  
  COUNT(DISTINCT customer_key) AS total_customers,  
  SUM(revenue) AS total_revenue,  
  ROUND( SUM(revenue) * 100.0 / SUM(SUM(revenue)) OVER (),2) AS revenue_percentage  
FROM age_category  
GROUP BY age_group  
ORDER BY total_revenue DESC;
```

Age Group	Total Orders	Total Customers	Total Revenue (M)	Revenue Percentage (%)
>60	21230	4100	\$18,86	34
41-60	18493	3733	\$16,30	29
26-40	14132	2893	\$12,61	23
18-25	7694	1626	\$6,80	12
12-17	1335	351	\$1,20	2

The business appears to rely significantly on older age segments (41+), which contribute more than half of total transactions.

Which sales channel contributes most effectively to overall performance?

```
SELECT
CASE
  WHEN st.store_key = 0 THEN 'Online'
  ELSE 'Offline'
END AS store_type,
COUNT(DISTINCT s.order_number) AS total_orders,
SUM(s.quantity * p.unit_price_usd) AS total_revenue,
ROUND(SUM(s.quantity * p.unit_price_usd)/COUNT(DISTINCT s.order_number),2) AS AOV
FROM sales s
JOIN stores st ON s.store_key = st.store_key
JOIN products p ON s.product_key = p.product_key
GROUP BY store_type
ORDER BY total_revenue DESC;
```



The offline channel leads overall performance, generating \$44.35M in revenue compared to \$11.40M online, with a slightly higher AOV (213.78 vs 204.38). The performance gap is mainly driven by transaction volume, suggesting online growth potential through scaling order frequency.

How are customers and sales distributed geographically?

```
SELECT
  c.continent,
  COUNT(DISTINCT c.customer_key) AS total_customers,
  COUNT(DISTINCT s.customer_key) AS active_customers,
  COUNT(DISTINCT s.order_number) AS total_orders,
  ROUND(SUM(s.quantity * p.unit_price_usd), 2) AS total_revenue,
  ROUND(
    COUNT(DISTINCT s.customer_key) * 100.0 /
    COUNT(DISTINCT c.customer_key),
    2
  ) AS cust_active_percentage,
  ROUND(
    SUM(s.quantity * p.unit_price_usd) * 1.0 /
    NULLIF(COUNT(DISTINCT s.order_number), 0),
    2
  ) AS AOV
FROM customers c
LEFT JOIN sales s
  ON c.customer_key = s.customer_key
LEFT JOIN products p
  ON s.product_key = p.product_key
GROUP BY c.continent
ORDER BY AOV DESC;
```

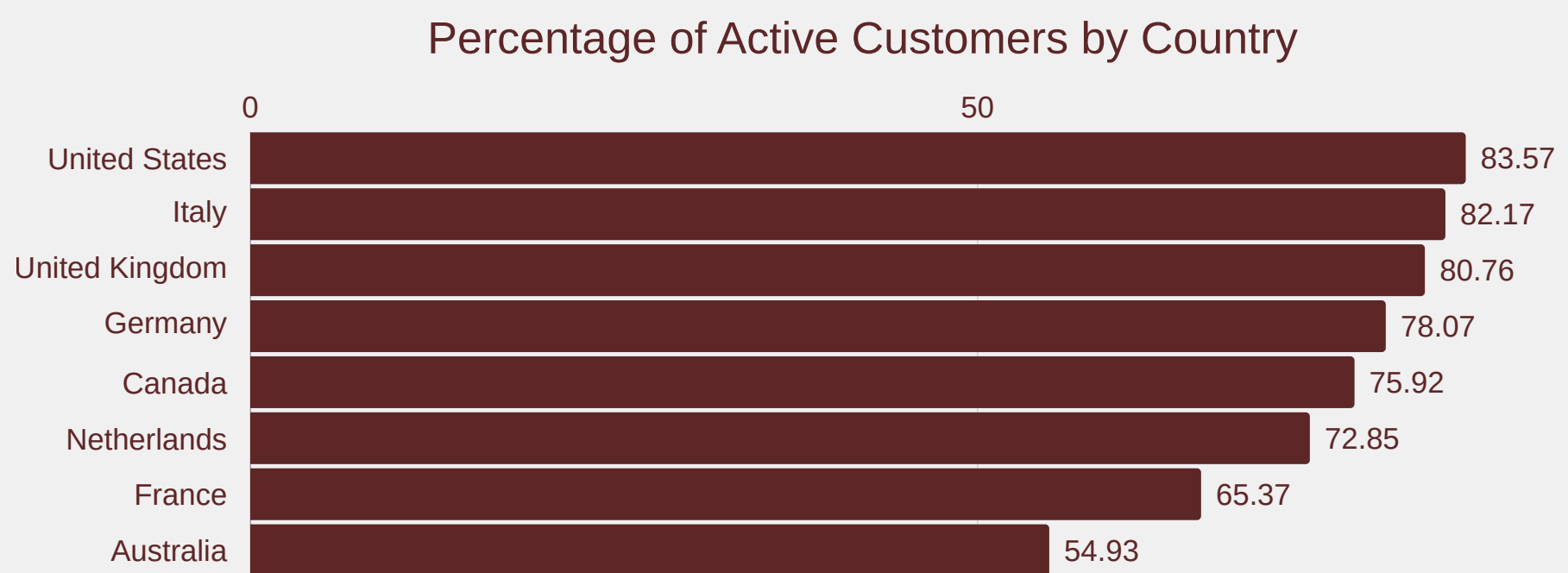
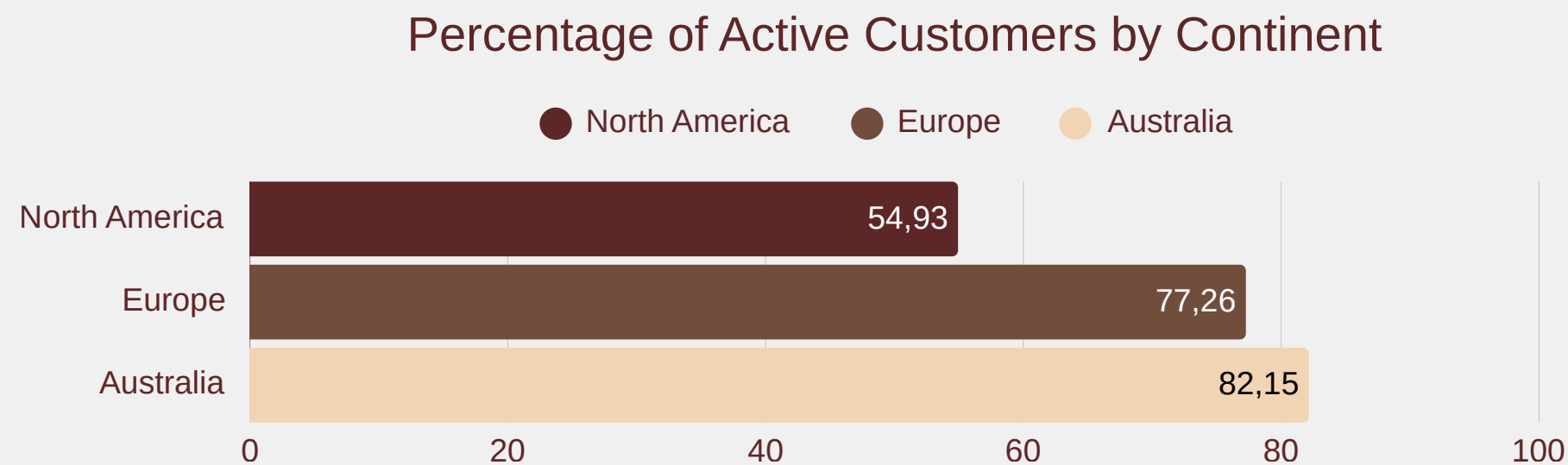
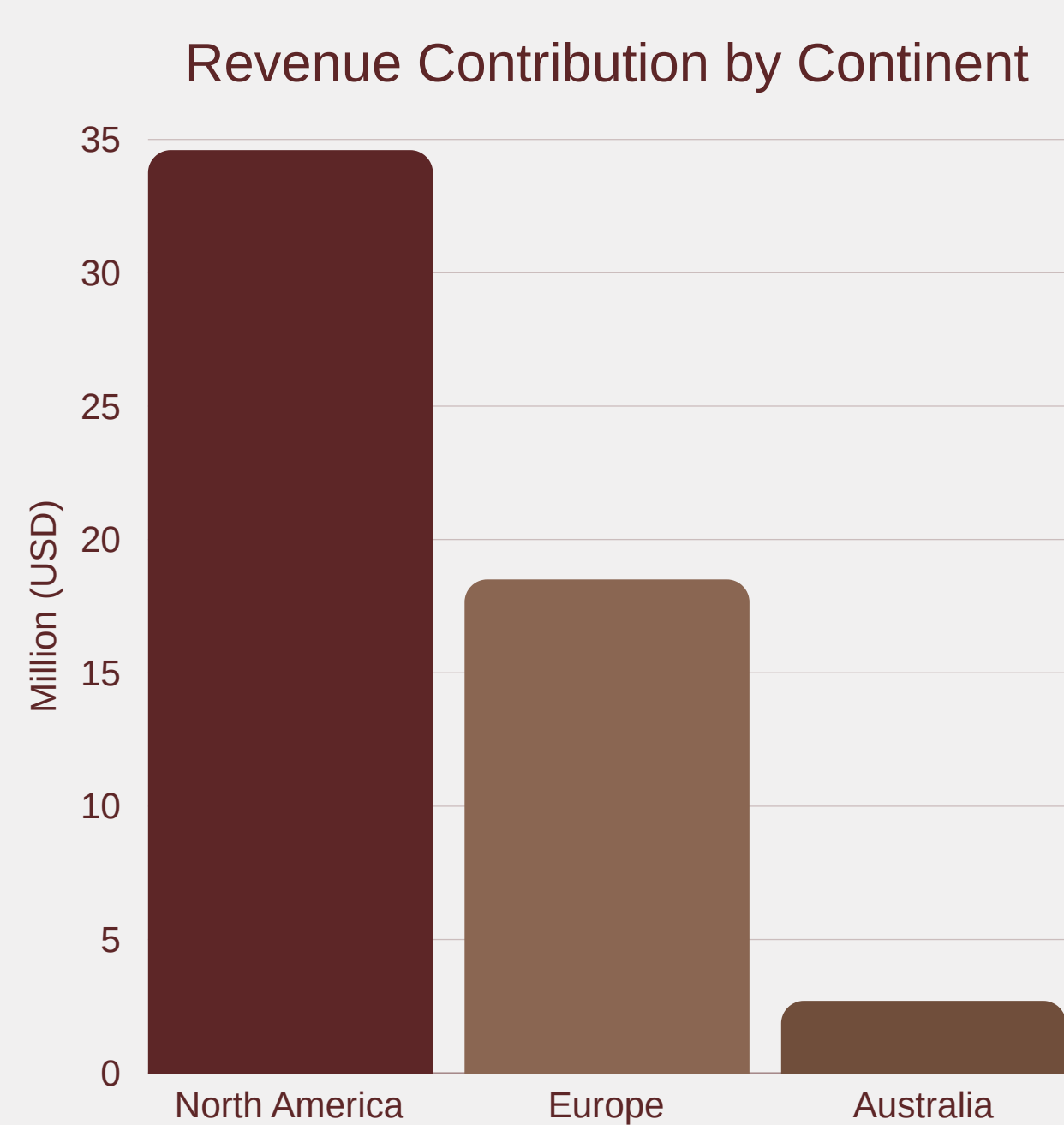
Continent	Total Customers	Active Customer	Total Orders	Total Revenue (USD)	Active Cutomers (%)	AOV
Australia	1420	780	1182	2.708.138	54,93	2.291,15
Europe	5465	4222	8642	18.451.376	77,26	2.135,08
North America	8381	6885	16502	34.595.966	82,15	2.096,47

North America leads in overall performance, with the highest customer activation rate (82%) and the largest revenue contribution. Strong engagement and high order volume are driving its dominance.

Europe shows solid and consistent performance, with a high activation rate (77%) and steady spending behavior, making it a stable contributor to total revenue.

Australia has the lowest activation rate (55%), but it records the highest AOV (2,291), meaning active customers tend to spend more per purchase, highlighting strong growth potential if engagement improves.

How are customers and sales distributed geographically?



North America dominates overall performance due to strong customer engagement and transaction volume, while Australia demonstrates the highest spending intensity per order, suggesting untapped revenue potential if customer activation improves.

BUSINESS INSIGHTS :

- ➔ Computers generate the largest share of revenue and profit, making it the core driver — but also a concentration risk. Diversifying growth across other categories can reduce volatility.
- ➔ The 2019–2021 decline suggests macro-level impact, while clear year-end peaks highlight strong seasonality. Adaptive cost control and demand-aligned planning are essential.
- ➔ More than half of total transactions come from customers aged 41+, indicating strong loyalty in mature segments but potential under-penetration in younger demographics.
- ➔ Offline remains the primary revenue driver, supported by higher transaction volume. However, since AOV is comparable, increasing online order frequency presents a strong growth opportunity.
- ➔ North America leads in revenue and customer activation, while Australia records the highest AOV despite lower engagement, indicating untapped high-value potential.



Thank you

If you have any suggestions or feedback, feel free to reach out.
I'm open to discussion and collaboration opportunities.



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